

GITIKA BABBAR

Cybersecurity Professional | SOC Analyst | Network Security Engineer

NY, USA | +1 (551)-235-5947 | gb2889@nyu.edu | [GitHub](#) | [Website](#) | [Linkedin](#)

SUMMARY

Results-driven Cybersecurity Professional with a Master's in Cybersecurity from NYU and 3 years of hands-on SOC and network security experience. Proven ability to detect, investigate, and respond to high-severity incidents across enterprise environments. Currently building AI-powered defensive security tools integrating Claude API with Python to automate SOC workflows including alert triage, threat hunt query generation, and executive report summarization. Actively sharpening threat detection skills through TryHackMe (Top 8% globally). Eager to contribute technical depth and a continuous learning mindset to a forward-thinking security team.

TECHNICAL SKILLS

Security Monitoring & IR: Splunk, Microsoft Sentinel, SOAR, Threat Detection, Incident Triage, Security Dashboards

Vulnerability Management: Qualys, Nessus, Metasploit, Nmap, Patch Management, Risk Assessments

IAM & Network Security: Zero Trust, MFA, Azure AD, RBAC, Firewalls, IDS/IPS, VPN, Hybrid Cloud Monitoring

Governance & Compliance: PCI DSS, SOX, ISO 27001, Root Cause Analysis, Risk Reporting

AI & Development: Python, Claude API, LLM Integration, JSON/CSV Processing, Security Workflow Automation

PROFESSIONAL EXPERIENCE

Infosys, India

Jan 2021-Dec 2023

SOC Analyst, Global Enterprise Clients, US Compliance Scope

- Resolved 150+ high-severity incidents annually phishing, brute-force, privilege escalation — driving 22% year-over-year SOC efficiency gain
- Tracked 80% critical vulnerability closure within SLA; reduced SLA breach rate 30% through cross-functional coordination and deadline enforcement.
- Reduced phishing recurrence 25% through payload analysis, SIEM rule tuning, and targeted security awareness reporting
- Improved threat detection coverage 20% via Metasploit/Nmap exploit simulations to close SIEM detection gaps
- Produced executive root cause analysis reports outlining incident scope, containment actions, and leadership recommendations

SECURITY PROJECTS

Python, Claude API, GitHub | Built to automate real SOC workflows experienced firsthand at Infosys

- SOC Alert Triage Tool:** Ingests security alerts from JSON/CSV (simulating SIEM exports), analyses severity and recommended actions via Claude API, generates timestamped analyst reports with interactive filtering menu.
- Threat Hunt Query Generator:** Converts plain English threat descriptions into platform-specific queries for Splunk, Microsoft Sentinel, and Elastic; maintains persistent query history for analyst reference.
- Executive Security Report Summarizer:** Transforms weekly incident data into C-suite ready reports with compliance framework references (MITRE ATT&CK, NIST, PCI DSS) and team performance metrics.
- Secure Chat App-TEXTNOW:** (Python, AES-GCM, ECDH, ECDSA) Built Signal-inspired encrypted chat application implementing AES-256-GCM confidentiality, ECDH key exchange, ECDSA authentication, and replay protection — with GUI interface and forensic message verification tools.

EDUCATION

Masters of Cyber Security | New York University, NY, USA

Jan 2024 - Dec 2025

Bachelors of Computer Science | Guru Gobind Singh Indraprastha University, India

Aug 2016 - Nov 2020

LEADERSHIP & ACTIVITIES

- Continuous Learning TRYHACKME:** Actively completing specialized paths in **SOC Level 1**, **Network Security** and **Microsoft Azure Security** to stay current with evolving threat landscapes.
- 2nd Place, US-Canada CSAW 2025 Operational Technology Security Challenge, NYU Tandon**
- Graduate Orientation Captain, NYU Tandon (Aug 2025):** Directed 40+ leaders to onboard 1,000+ graduate students; managed logistics, academic integration, and community-building initiatives.
- Teaching Fellow, Applied & Post-Quantum Cryptography (Fall 2025):** Supported 60+ graduate students; facilitated labs, research projects, and GitHub-based implementations connecting theory with practical cryptographic methods.
- Course Assistant, Cybersecurity K12 Program (Summer 2025, Summer 2026):** Delivered Linux, cryptography, and digital forensics activities for 120+ high school students; mentored on coding and cybersecurity career pathways, AI Practical learnings.
- Marketing Coordinator, ISAB (Fall 2024 - Fall 2025):** Led outreach campaigns for 500+ international students; promoted resources and student engagement initiatives.
- Volunteer, Bsides NYC (Oct 2025):** Supported logistics, speaker coordination, and attendee engagement at a leading cybersecurity community conference.
- WiCyS Mentee, 2025 Cohort:** Selected for competitive mentorship; engaged in chapter events, virtual conferences, and career fairs.
- Sponsored Attendee, SWE Conference 2025:** Awarded full sponsorship recognizing leadership and STEM advocacy.